

Problem statement 1:

Problem Statement:

Given a positive integer N , print all the numbers from 1 to N in increasing order, separated by a single space.

Your task is to write a program that prints the required sequence.

After completing the program, analyze your solution by calculating its Time Complexity and Space Complexity, and explain why your answer is correct.

Input Format

A single integer N .

Output Format

Print all integers from 1 to N in increasing order separated by spaces.

Constraints

$1 \leq N \leq 10^5$

Example 1

Input

5

Output

1 2 3 4 5

Explanation

The numbers from 1 to 5 are printed in increasing order.

Example 2

Input

8

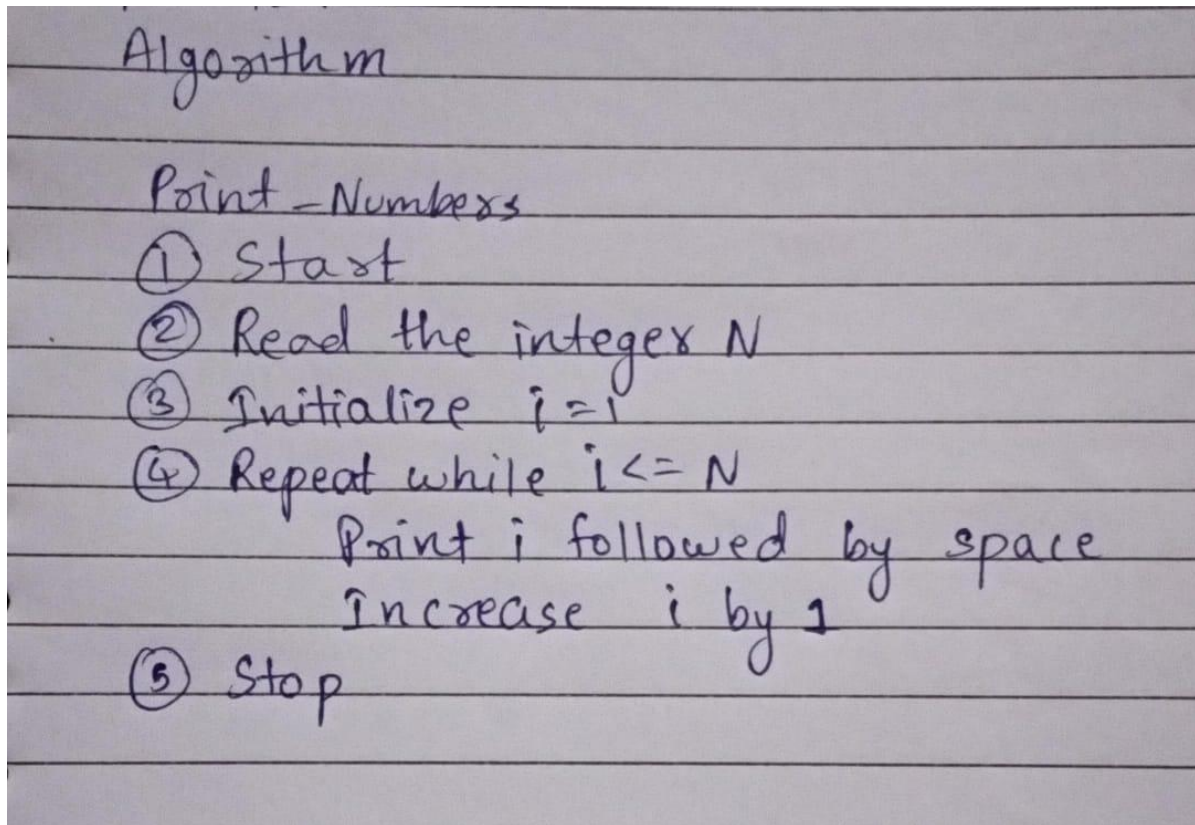
Output

1 2 3 4 5 6 7 8

Explanation

The program prints every integer starting from 1 until 8, maintaining increasing order.

Algorithm:



Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int N;
7      std::cin >> N;
8
9      for (int i = 1; i <= N; i++)
10     {
11         cout << i << " ";
12     }
13
14     return 0;
15 }
16
```

Output:

```
PS C:\Users\priya\OneDrive\Desktop\Week 4 Module> .\Problem_1.exe
5
1 2 3 4 5
```

Complexity:

Complexity

Time -
 $O(N)$

- The loop executes exactly ~~N~~ N times, once for each number from 1 to N

Space -
 $O(1)$

- The program only uses a fixed number of variables (N and i). The amount of memory used does not ~~represent~~ increase with N

Problem statement 2:

Problem Statement:

Given a positive integer N, find the sum of the first N natural numbers using a loop.

Your task is to write a program that calculates and prints the required sum.

After completing the program, analyze your solution by calculating its Time Complexity and Space Complexity, and explain your answer.

Input Format

A single integer N.

Output Format

Print the sum of the first N natural numbers.

Constraints

$$1 \leq N \leq 10^7$$

Example 1

Input

5

Output

15

Explanation

The sum of the first five natural numbers is:

$$1 + 2 + 3 + 4 + 5 = 15$$

Example 2

Input

10

Output

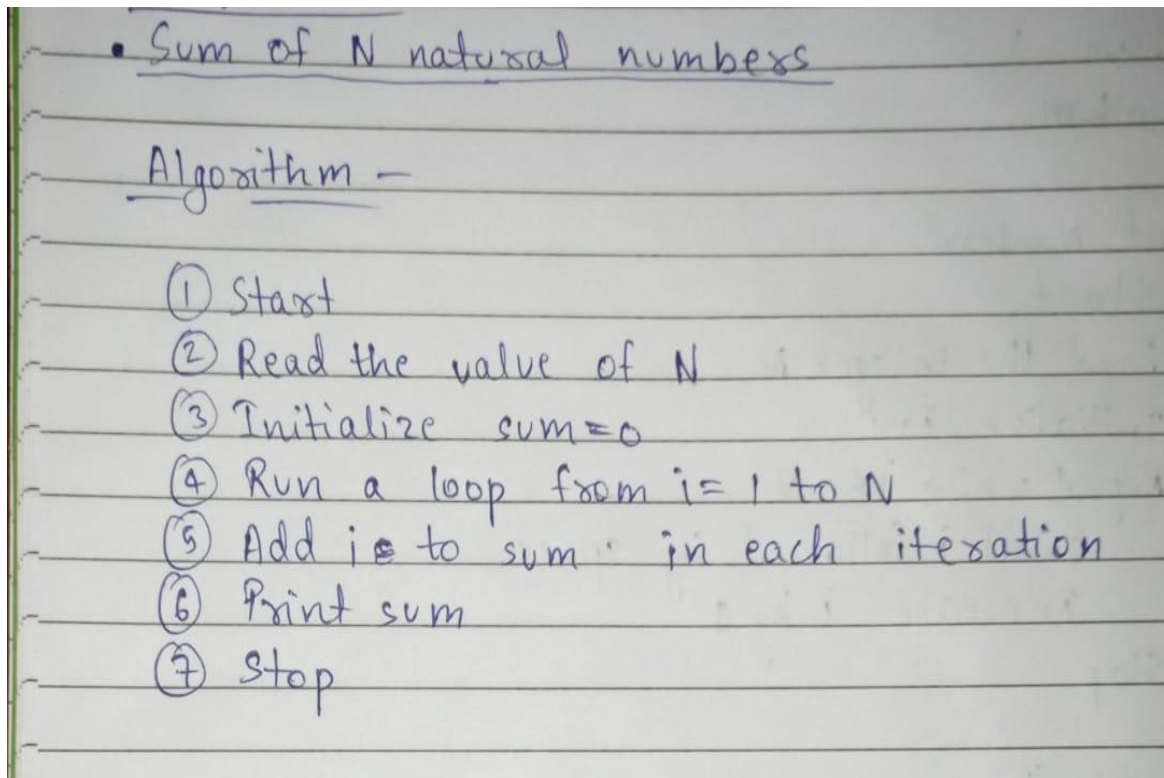
55

Explanation

The sum of the first ten natural numbers is:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

Algorithm:



Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int N;
7      long long sum = 0;
8
9      cin >> N;
10
11     for (int i = 1; i <= N; i++)
12     {
13         sum = sum + i;
14     }
15
16     cout << sum << endl;
17
18     return 0;
19 }
```

Output:

```
PS C:\User
5
15
PS C:\User
8
36
PS C:\User
```

Complexity:

Complexity —

- Time — $O(N)$
Execution increases with N
as loop repeats itself N times.
- Space — $O(1)$
Fixed no. of variables (N, i, sum) are
used regardless of N 's value

Problem Statement 3 :

Problem Statement:

Given two integers A and B, calculate the value of A^B (A raised to the power B) using a loop.

Your task is to write a program that computes the result and prints it.

Once your program is complete, analyze your solution by calculating its Time Complexity and Space Complexity, and explain why your answer is correct.

Input Format

Two integers A and B separated by a space.

Output Format

Print the value of A^B .

Constraints

$$1 \leq A \leq 100$$

$$0 \leq B \leq 20$$

Example 1

Input

2 5

Output

32

Explanation

The value of 2^5 is:

$$2 \times 2 \times 2 \times 2 \times 2 = 32$$

Example 2

Input

3 4

Output

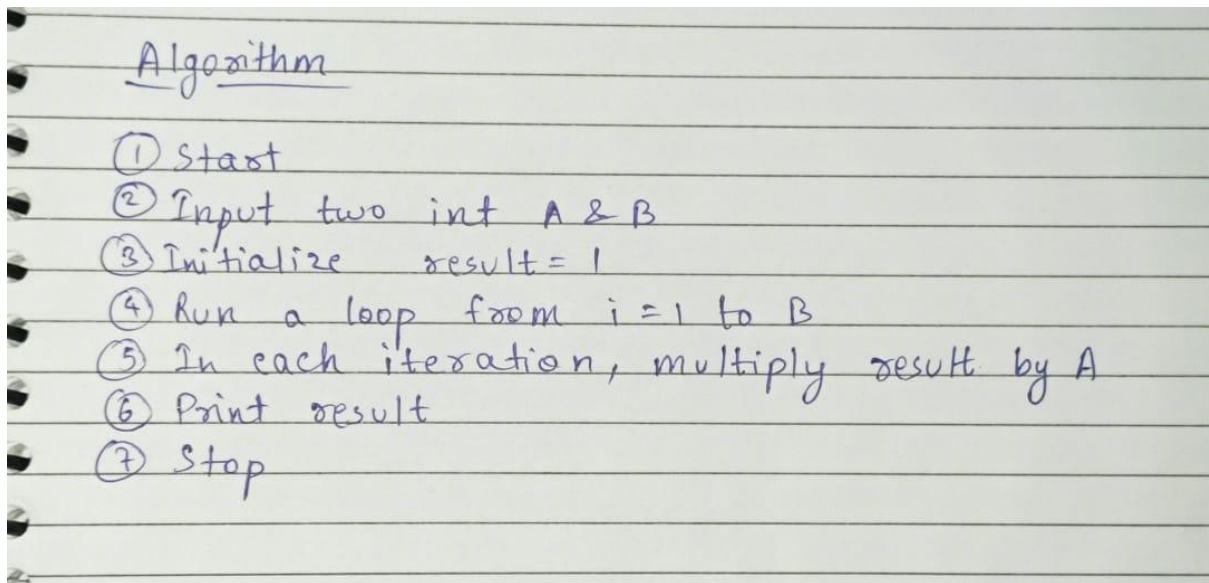
81

Explanation

The value of 3^4 is:

$$3 \times 3 \times 3 \times 3 = 81$$

Algorithm:



Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      long long A, B;
7      long long result = 1;
8
9      cin >> A >> B;
10
11     for (int i = 1; i <= B; i++)
12     {
13         result = result * A;
14     }
15
16     cout << result << endl;
17
18     return 0;
19 }
```


Output:

```
PS C:\Users\ [redacted] \OneDrive\Desktop\Week 4 Module> .\Num_Sq.exe
5
5
3125
PS C:\Users\ [redacted] \OneDrive\Desktop\Week 4 Module> .\Num_Sq.exe
4
5
1024
PS C:\Users\ [redacted] \OneDrive\Desktop\Week 4 Module> .\Num_Sq.exe
2
8
256
PS C:\Users\ [redacted] \OneDrive\Desktop\Week 4 Module> |
```

Complexity:

Complexity -

- Time - $O(B)$
The loop executes B times. In every iteration, one multiplication is performed, so the runtime increases with increasing no. of B .
- Space - $O(1)$
The program uses only a fixed no. of variable i.e. A, B, i and result. No additional memory is allocated based on input size.